

Wizard's Chess System Specification

Comprehensive Technical Architecture for Voice-Controlled Hybrid Physical & 3D Chess

PROJECT SCOPE

Full System Architecture

INTERFACE PARADIGM

Voice & Electro-Mechanics

DOCUMENT VERSION

v2.0 • Expanded Specifications

1. System Executive Overview

The Wizard's Chess system is a dual-interface chess platform combining real-time speech recognition, 3D animated web rendering, and an automated mechatronic physical chessboard. Players issue spoken move commands (e.g., "Knight to F3"), which are processed through a phonetic normalization layer and validated by an engine core before translating into simultaneous physical motor operations and animated visual rendering.

SYSTEM DATA FLOW PIPELINE

[Voice Input] → [Web Speech API / Parser] → [chess.js State Engine]

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[3D Web UI: Three.js + GSAP] [Hardware: WebSockets / Serial]
├ Piece Arc Interpolation ─ Microcontroller (ESP32/Arduino)
├ Shatter/Dust Particle Effects ─ CoreXY Stepper Gantry Movement
└ Audio FX & Voice Feedback ─ 12V Electromagnet & LED Grid

2. Voice Interface & Command Disambiguation

Voice recognition presents challenges regarding phonetic misinterpretation of chess algebraic coordinates. The speech recognition pipeline utilizes a fuzzy phonetic parser to map continuous audio transcriptions into standardized SAN notation.



Speech Recognition

Uses Web Speech API or local Whisper models. Captures continuous audio streams, returning raw transcriptions for command evaluation.



Phonetic Mapping

Resolves ambiguous terms: "Night" → Knight, "See" → C-file, "Rock/Rook" → Rook, "To/Two/Too" → 2. Normalizes phonetic overlaps.



Legal Move Validation

Executes logic via chess.js. Cross-references parsed commands against all legal moves for the current board state.



Audio Disambiguation

If a command is ambiguous (e.g., "Rook to d1" with two valid rooks), voice synthesis prompts: "Which rook to d1?"

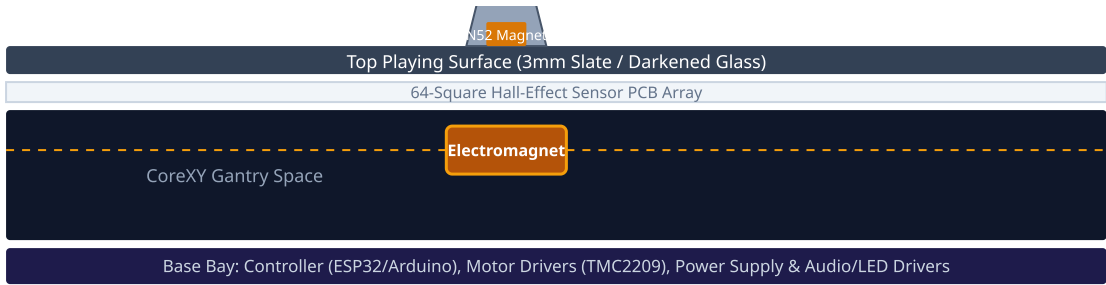
Command Grammar Standard

Accepts flexible phrasing variants: "Knight to f3", "Pawn e2 to e4", "Pawn takes d5", and "Castle kingside". Invalid moves trigger spoken error responses without altering board state.

3. Physical Board Mechatronics Engineering

The physical board utilizes a silent, under-board CoreXY gantry mechanism. An electromagnet drives carved marble pieces equipped with embedded neodymium bases without visible external mechanisms.

Physical Enclosure Cross-Section



4. Hardware Subsystem Specifications

Subsystem	Component Selection	Functional Role
Motion Mechanism	CoreXY Belt Frame + Dual NEMA 17 Steppers	Provides smooth X and Y positioning under the board grid using silent stepper drivers.
Magnetic Coupling	12V 5kg Holding Force Electromagnet	Grips the N52 neodymium base of pieces through the board during movement sequences.
State Detection	8x8 Hall-Effect Sensor Matrix Array	Detects piece presence across all 64 squares for manual setup and move verification.
Microcontroller	ESP32 / Raspberry Pi Pico	Receives target coordinates from web app over Serial/WebSocket and executes motor paths.
Feedback Systems	RGB Addressable LEDs & 10W Sound Transducer	Provides visual flares beneath target squares and physical stone clatter audio feedback.

5. Pathfinding & Seam Routing Algorithm

To navigate pieces across occupied tiles without collisions, the software executes pathfinding along the seam lines separating the squares (0.5 -square offset alignment):

- Step 1 (Engagement):** The electromagnet homes beneath the piece and energizes.
- Step 2 (Offset Alignment):** The piece is shifted 0.5 squares onto the nearest seam line.
- Step 3 (Traversal):** The piece moves along grid seams around occupied squares to its destination coordinate.
- Step 4 (Settling):** The piece aligns centered on the target square, and the electromagnet de-energizes.

6. Movie-Accurate Harry Potter Piece Architecture

Replicating McGonagall's Chamber set from *Harry Potter and the Philosopher's Stone* (inspired by the 12th-century Isle of Lewis chessmen). Each piece features bespoke physical sculpts, custom mass distribution, and embedded magnetic bases to ensure physical motion stability.

 King Sculpt: Seated monarch on throne holding a unsheathed broadsword. Height / Weight: 130mm / 185g Core Magnet: Dual N52 Disc (20mm x 5mm) Base Pad: Low-friction PTFE film	 Queen Sculpt: Crowned figure holding a scepter, leaning in an aggressive posture. Height / Weight: 120mm / 165g Core Magnet: Single N52 Disc (25mm x 5mm) Base Pad: Polished PTFE base ring	 Bishop Sculpt: Robed cleric with miter helm holding a staff in a forward stride. Height / Weight: 105mm / 130g Core Magnet: Single N52 Disc (18mm x 4mm) Base Pad: PTFE felt compound
 Knight Sculpt: Armored warrior on a rearing warhorse (Ron's canonical piece). Height / Weight: 115mm / 155g Core Magnet: High-Torque N52 Dual Core Base Pad: Low-friction PTFE film	 Rook (Castle) Sculpt: Living stone sentry wielding a double-handed war hammer. Height / Weight: 100mm / 145g Core Magnet: Single N52 Disc (20mm x 5mm) Base Pad: Wide-perimeter PTFE pad	 Pawn Sculpt: Kneeling medieval infantry with round shield and short broadsword. Height / Weight: 70mm / 80g Core Magnet: Single N52 Disc (15mm x 3mm) Base Pad: Polished PTFE disc

7. Piece Manufacturing & Magnetic Balance Matrix

Piece Type	Material Composition	Physical Base Diameter	Electromagnet Pull Force (12V)
King & Queen	Cast Stone Resin / Cold-Cast Bronze Core	42mm Base Diameter	4.8 kgf holding force required
Knight & Rook	Cast Marble Powder Aggregate	38mm Base Diameter	4.2 kgf holding force required
Bishop	Synthetic Marble Resin	34mm Base Diameter	3.5 kgf holding force required
Pawn	High-Density Polymer Resin	28mm Base Diameter	2.8 kgf holding force required

Center of Gravity & Magnet Placement Specification

To prevent tall pieces (e.g., King & Knight) from tipping during high-speed Gantry accelerations, internal brass counterweights are embedded into the lower 20% of the piece bodies, maintaining a low center of gravity directly over the N52 base magnet.

8. 3D Web App Visuals & Harry Potter Capture Shaders

The web application renders real-time 3D mirrors of the physical pieces using Three.js and React Three Fiber. Captures trigger theatrical destruction animations inspired by the movie scene.

3D PIECE ATTACK & SHATTER ANIMATION LOOP

1. ATTACK CHARGE —► Attacking 3D piece raises off board (Y-axis arc) and charges target square. 2. STRIKE IMPACT —► Weapon strike event (e.g., Rook hammer smash or Queen sword sweep). 3. FRAGMENTATION —► Target model swaps to pre-fractured Voronoi mesh; pieces explode with physics. 4. DUST & FADE —► Smoke/dust particle shader spawns while fragments dissolve into the board. 5. PHYSICAL SYNC —► Under-board electromagnet drags physical piece off-grid into graveyard lane.

9. Piece-Specific Destruction FX Specs

Attacker	Harry Potter Combat Motion	Web3D Shader & Particles	Physical Audio SFX
Queen	360° spinning scepter slash destroying adjacent piece.	Golden light arc particle trail + stone debris spray.	Sharp metallic blade ring + heavy stone crash.
Rook	Overhead double-handed hammer slam smashing target.	Ground shockwave ring shader + crumbling Voronoi mesh.	Deep thundering rumble + explosive stone impact.
Knight	Horse rears back and tramples target with hooves.	Dust cloud particle emitter + debris scatter force.	Horse whinny + heavy hoof stone strikes.
Bishop	Staff thrust piercing the heart of the target piece.	Energy pulse streak along staff + fracturing mesh.	Resonant magical hum + crumbling stone crumble.

10. Implementation Roadmap

Phase	Deliverable	Key Milestones
Phase 1	Voice Engine & 3D Web App	Set up Web Speech parser, chess.js state engine, and Three.js 3D board rendering with move animations.
Phase 2	Gantry Mechanics & Firmware	Assemble CoreXY frame, write ESP32 G-code stepper pathfinder, and test electromagnet coupling force.
Phase 3	Sensory Matrix & Integration	Integrate 8x8 Hall-effect array, establish WebSocket bridge between web app and hardware, and tune audio/LED FX.
Phase 4	Enclosure & Movie Piece Fabrication	Cast/3D-print HP-accurate pieces, embed N52 magnets into bases, apply PTFE pads, and execute system integration testing.

Specification Sign-off

This complete 4-page design document establishes the baseline for voice parsing, mechatronic gantry design, Harry Potter piece fabrication, and Web3D visual effects.

